Industrial Decarbonisation Pathway Analysis

Calculation Methodology

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Reference document for senior reviewers and technical assurance

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Document control

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1. Introduction

1.1 Purpose

This document sets out the calculation methodology underpinning the analysis system described herein (referred to throughout as *the Engine*). The Engine produces consultancy-grade decarbonisation pathway studies for industrial sites, focused in the present version on steam-system electrification within the food and drink, chemicals, pharmaceutical, paper and refining sectors.

The methodology is published so that a senior chartered engineer reviewing an Engine output can, within approximately one hour, verify that: (a) every numeric output traces to a deterministic calculation against a recognised standard or peer-reviewed source; (b) the assumptions, default values and uncertainty treatments are defensible and documented; (c) the limitations of the analysis are stated explicitly and have been respected.

1.2 Scope of the Engine

The Engine accepts an industrial site brief — sector, sub-sector, energy profile, plant inventory and operational constraints — and produces a multi-year decarbonisation pathway covering technology screening, dispatch simulation, investment sequencing, uncertainty quantification, safety screening, grid connection feasibility and reliability assessment. The output deliverable is a written technical report supported by a complete calculation provenance record.

The Engine does not generate detailed engineering design (single-line diagrams, P&IDs, control narratives, foundation calculations or hazardous-area drawings). It is a feasibility-and-pathway tool, sized to inform senior-engineer review at the Concept Design and Feasibility stages of a typical RIBA or equivalent design programme.

1.3 Intended audience

This document addresses two audiences in parallel.

The first is the **senior consulting engineer** receiving an Engine output as a draft. The methodology gives that engineer the visibility required to challenge any number in the report, reproduce its derivation and decide whether to sign the deliverable in its entirety, in part or with revision.

The second is the **technical assurance reviewer**, including those acting on behalf of professional indemnity insurers. The methodology gives that reviewer the audit trail required to confirm that every calculation in scope is traceable to a recognised standard, that no critical assumption is implicit, and that the boundary between machine-generated content and human professional judgement is unambiguous.

1.4 Document structure

Section 2 describes the Engine's architecture and the design principles that govern every module within it. Section 3 sets out the calculation methodology of each module, with inputs, procedure, outputs and standards cited. Section 4 describes the provenance, validation and quality-assurance regime. Section 5 declares the limitations of the methodology. Section 6 contains a consolidated register of standards and references.

2. System Architecture

2.1 Overall design

The Engine consists of three components: a deterministic simulation and optimisation engine written in Python, a knowledge corpus of standards, manufacturer datasheets and case-study evidence stored under retrieval, and an orchestration layer based on a large language model.

The deterministic engine is the locus of every numerical claim made in the deliverable. It is independently runnable: each module can be invoked from a Python interpreter without the orchestration layer present. The knowledge corpus supplies citations and evidence to support the qualitative narrative of the deliverable; it does not perform calculation. The orchestration layer plans the sequence of analysis steps, invokes the engine modules in the correct order, retrieves supporting evidence from the corpus and writes the narrative around the engine's numerical results. It does not perform arithmetic, and is constrained against doing so.

2.2 Non-negotiable design principles

Five principles govern every module of the Engine without exception.

(i) No arithmetic in the orchestration layer. Every numeric claim in the output, without exception, traces to a deterministic Python function call in the engine. The orchestration layer is permitted to interpret and contextualise numbers, but not to compute them. This constraint is enforced both by tool-call architecture and by self-check at output time.

(ii) Calculation provenance in every output. Each numeric output is accompanied by a record naming the function that produced it, the input values used, the version of the engine and the standard or reference document underpinning the calculation. The provenance record is part of the output schema; it is not optional.

(iii) The engine is independently runnable. The Python engine produces the same numbers irrespective of whether the orchestration layer is present. This requirement guards against calculation drift and supports independent re-execution by a reviewer.

(iv) Self-critique loop. Before producing the final deliverable, the orchestration layer reviews its own draft against a checklist of common inconsistencies — figures that disagree across sections, claims unsupported by tool calls, assumptions undeclared. Findings are addressed before output.

(v) Golden test cases gate every release. No engine module is released until it produces results, against three reference site profiles, that fall within tolerance bands set by hand-checked engineering judgement. The reference profiles are dairy (5 MW), brewery (8 MW) and soft drinks (12 MW).

2.3 Data flow

A site brief is parsed into a structured energy profile (Module 3.1) from which a baseline carbon trajectory is computed (3.2). The technology screen (3.5) produces a shortlist of feasible options. For each candidate pathway, the dispatch simulator (3.3) runs an 8,760-hour operational simulation, drawing on refrigerant-cycle thermodynamics (3.4) for any heat pump components. The investment pathway optimiser (3.6) selects the preferred sequencing under techno-economic and carbon objectives. The Monte Carlo module (3.7) propagates uncertainty across price, demand and grant-outcome inputs. Heat integration is assessed via pinch analysis (3.8). Safety, grid connection and reliability constraints (3.9–3.11) are checked against the proposed configuration. The orchestration layer composes the resulting numerical record, with citations from the knowledge corpus, into the deliverable.

3. Calculation modules

3.1 Energy profile parsing

The procedure constructs a half-hourly energy profile (17,520 timesteps per year) for each declared end-use — process steam, hot water, space heating, process cooling, motive electricity, lighting and compressed air — broken down by fuel and by production-volume linkage. Where half-hourly metering data is supplied, the profile is constructed directly from it; otherwise, sector-specific shape templates are scaled to the declared annual consumption and overlaid with the supplied production schedule.

The procedure produces a load duration curve per end-use, a base-load / variable-load split, a production-linkage coefficient (kWh per tonne, hectolitre or unit produced) and an asset-by-asset utilisation factor for the existing plant inventory. Identified inefficiencies are flagged against published sector benchmarks (ETSU industrial sector benchmarking; BREF documents).

Standards cited: *BS EN 16247-1, BS EN 16247-3, CIBSE TM54.*

3.2 Baseline carbon accounting

Baseline emissions are computed in compliance with the GHG Protocol Corporate Standard, using DEFRA UK Government GHG Conversion Factors for the year of analysis. Scope 1 combustion emissions are reported with separate carbon dioxide, methane and nitrous oxide components, in line with SECR reporting expectations. Scope 2 electricity emissions are computed both location-based (using half-hourly grid carbon intensity from the National Energy System Operator's Future Energy Scenarios dataset) and market-based (where REGOs or PPAs are declared). Scope 3 upstream natural gas emissions, including methane leakage and production-stage emissions, are added explicitly; this term is typically material, in the range 15–20% of Scope 1.

The procedure also flags exposure to the Climate Change Levy, the UK Emissions Trading Scheme (where the site exceeds the 20 MW combustion threshold or operates a regulated activity) and the UK Carbon Border Adjustment Mechanism (where applicable to product exports).

Standards cited: *GHG Protocol Corporate Standard and Scope 2 Guidance; DEFRA GHG Conversion Factors and Methodology Paper; UK ETS Order; CCL rates schedule.*

3.3 Site dispatch simulation

The dispatch simulator advances the site through 8,760 hourly timesteps (with a half-hourly mode where electricity tariff or grid intensity resolution requires it), allocating heat and electricity demand across the proposed technology stack at each step. Multi-pressure steam headers are tracked separately, with saturated-steam properties resolved against IAPWS-IF97 formulations. Heat pump coefficient of performance at each timestep is computed by the refrigerant-cycle module (3.4) at the actual source and sink temperatures prevailing at that step; Carnot or seasonal-average approximations are not used.

The simulator supports four dispatch policies: merit-order (cost-minimal at short-run marginal cost), carbon-minimal (lowest emissions per kilowatt-hour delivered), Pareto-weighted (configurable cost / carbon trade-off) and regulatory-constrained (subject to ETS allowance, CCA cap or planning use-class limits). Equipment ramp rates, minimum runtimes and minimum downtimes are respected. An energy balance check is enforced at module exit and the routine raises an exception where the closure error exceeds 0.5%.

Standards cited: *CIBSE AM17, CIBSE TM54, IChemE process integration good practice, NESO Future Energy Scenarios.*

3.4 Refrigerant cycle thermodynamics

The heat pump cycle is solved against a published equation-of-state via CoolProp, supporting the refrigerants commonly encountered in industrial high-temperature applications: ammonia (R717), carbon dioxide (R744, including transcritical operation), R1234ze(E) and selected legacy fluorocarbons where their use is presently lawful under retained F-Gas regulation.

For each operating point, the procedure resolves evaporator and condenser saturation states (with declared subcool and superheat), the compressor isentropic efficiency from a compressor-type-specific map (screw, piston, centrifugal), the discharge temperature (against refrigerant-specific limits) and the cycle-average coefficient of performance for both heating and cooling duties. The discharge temperature check is enforced; configurations that exceed the safe envelope are returned with a warning rather than a coefficient-of-performance value. Multi-stage, economiser, intercooled and cascade cycles are supported in the v0.3 release.

Standards cited: *BS EN 14825, BS EN 378 (all parts), F-Gas Regulation 517/2014 (UK retained), CoolProp documentation citing the underlying equations of state.*

3.5 Technology screening

The screening procedure evaluates each technology in the longlist against nine feasibility axes: thermodynamic compatibility (can the technology deliver the required temperature at a viable coefficient of performance), commercially available capacity range, refrigerant safety (BS EN 378 charge limits at the proposed location, F-Gas eligibility, ATEX zone implications), process compatibility (Good Manufacturing Practice constraints in food and pharmaceutical applications, contamination risk for biomass), grid headroom against existing connection capacity, planning consent and Building Regulations Part L implications, site footprint, compressor envelope and broader regulatory eligibility (UK ETS, CCA).

The output includes a shortlist with feasibility rationale and flagged risks per technology, an excluded list with reasoned exclusions referenced to the failing axis, and notes flagged for senior review where the screen has insufficient information to decide.

Standards cited: *BS EN 378, BS EN 14825, F-Gas Regulation 517/2014, DSEAR 2002, BS 7671, Approved Document L, Town and Country Planning (Use Classes) Order 1987.*

3.6 Investment pathway optimisation

The optimiser sequences capital investment over a 15- to 20-year planning horizon. In the present release, the procedure operates by enumeration: approximately fifty candidate pathways, each defined by a sequence of technology installations and capacity steps, are simulated end-to-end through the dispatch module (3.3) and ranked by net present value at the declared discount rate. A future release will replace enumeration with a multi-period stochastic mixed-integer linear programme implemented in Pyomo or OR-Tools, retaining the dispatch simulator as the inner-loop evaluator.

The procedure produces three named pathways — Conservative, Balanced and Aggressive — and a Pareto frontier of approximately ten alternatives across the cost-versus-carbon trade-off. For each pathway the output includes year-by-year capital expenditure against the declared budget envelope, simple and discounted payback, internal rate of return, levelised cost of heat, and risk-adjusted net present value computed at the 90% conditional Value-at-Risk. Sensitivity to electricity price, gas price and grant outcome is reported.

Standards cited: *HM Treasury Green Book; BS EN 16247-1 §6 (techno-economic appraisal); IEA Cost & Performance Database.*

3.7 Monte Carlo uncertainty

Uncertainty is propagated by Latin hypercube sampling, with default 1,000 trials and 10,000 trials available for high-stakes runs. The uncertain inputs are electricity and gas price trajectories (typically triangular distributions calibrated to DESNZ projections), grid carbon intensity in the 2030 horizon (NESO Future Energy Scenarios range), heat pump capital cost (manufacturer ranges), grant outcomes (Bernoulli on declared grant probability) and annual demand growth. Gas and electricity prices are sampled with a positive correlation, default 0.6, applied through a Gaussian copula.

Outputs include the full distribution of net present value (P10, P50, P90, mean, standard deviation, skew), the carbon trajectory uncertainty cone, the probability that net present value exceeds zero, the probability that an annual carbon target is met, and Value-at-Risk and conditional Value-at-Risk at the 95th percentile. Sobol first- and second-order sensitivity indices identify which inputs drive output variance; Morris elementary effects are reported as a screening-level cross-check.

Standards cited: *Saltelli et al., Global Sensitivity Analysis: The Primer; BS EN 16247-3 §6.4; HM Treasury Green Book §5.*

3.8 Pinch analysis and heat integration

Pinch analysis is performed in accordance with the Linnhoff March method as described in Kemp (Pinch Analysis and Process Integration, second edition) and IChemE process-integration practice. Hot and cold process streams are aggregated into composite curves and the grand composite curve. The pinch temperature, minimum hot utility demand, minimum cold utility demand and minimum heat-exchanger area target are reported at the user's declared minimum approach temperature (default 10 K). The capital-energy trade-off is presented as a curve over a range of approach temperatures.

A heat-exchanger network synthesis routine proposes stream-to-stream matches and provides a shell count and area estimate. Pinch violations and network pockets are flagged. The procedure does not perform detailed thermal-hydraulic design of individual exchangers; that is reserved for the subsequent design stage.

Standards cited: *Kemp, Pinch Analysis and Process Integration, 2nd ed.; Linnhoff March, User Guide on Process Integration; Smith, Chemical Process Design and Integration; IChemE Process Integration Best Practice Guides.*

3.9 Safety constraint screening

The safety screen is positioned at the level of an ALARP screening exercise — sufficient to identify the dominant safety considerations, insufficient to substitute for a HAZOP. For each proposed unit, the procedure performs a charge-limit assessment under BS EN 378 against the proposed plant-room volume; identifies refrigerants that trigger ATEX zone classification (with the typical 20–40% capital cost uplift for compliant equipment); assesses refrigerant-specific hazards (ammonia toxicity and leak detection requirements; carbon dioxide asphyxiation in confined space; hydrocarbon flammability); and screens for Pressure Equipment Directive applicability where the operating pressure exceeds 0.5 barg.

Steam-system safety implications under PSSR 2000 are flagged for new electrode boilers. DSEAR-relevant cause-event-consequence pairs are recorded. CDM 2015 construction-phase risks are noted. The output is a structured risk register intended to inform, not replace, the formal HAZOP that will be required at the subsequent design stage.

Standards cited: *BS EN 378 (parts 1–4); DSEAR Regulations 2002; ATEX Directive 2014/34/EU; F-Gas Regulation (UK retained); Pressure Equipment (Safety) Regulations 2016; PSSR 2000; CDM Regulations 2015.*

3.10 Grid connection assessment

The grid connection screen is positioned at the level of a G99 connection feasibility study, sufficient to identify site-killing connection issues but not to substitute for the Distribution Network Operator's formal connection offer. The procedure compares proposed new electrical loads (heat pumps, electrode boilers, motors) against the declared existing connection capacity. Where the resulting site demand exceeds the existing capacity, the increment requiring DNO reinforcement is reported, with an indicative cost range (£50,000 – £500,000) and an indicative timeline (six to eighteen months) calibrated to the relevant DNO area.

Voltage rise at the point of connection is estimated against BS EN 50160 limits. Harmonic injection from variable-speed drives and electrode boilers is screened against IEEE 519 and Engineering Recommendation G5/5; where indicative harmonic distortion exceeds the tolerance, a formal harmonic study is recommended.

Standards cited: *Engineering Recommendation G99; Engineering Recommendation G5/5; BS EN 50160; BS 7671; IEEE 519; DNO Long Term Development Statements (latest editions).*

3.11 Reliability and availability

System availability is computed from per-unit Mean Time Between Failures and Mean Time To Repair data, drawn either from vendor documentation or, in its absence, from the Offshore Reliability Data Handbook and Lees' Loss Prevention in Process Industries. Single-unit and N+1 / N+2 redundant configurations are evaluated against a user-declared availability target (typically 99.0%, 99.5% or 99.9%). The procedure reports annual expected downtime in hours, annual expected downtime cost at the user's declared cost-per-downtime-hour rate, and the recommended planned-maintenance regime.

A v0 implementation assumes exponential failure distributions and time-independent availability; future releases will support Markov modelling for systems with significant maintenance-state structure.

Standards cited: *BS EN 60300; OREDA (Offshore Reliability Data Handbook); Lees, Loss Prevention in the Process Industries; IEC 61078 (reliability block diagrams).*

4. Provenance, validation and quality assurance

4.1 Calculation provenance

Every numeric output of every module is accompanied by a provenance record naming the engine version, the module function, the input values supplied, and the reference document or standard underpinning the calculation. The provenance record is generated mechanically and is part of the output schema. A reviewer reading any deliverable can, for any number in the report, locate the originating function call and reproduce it.

4.2 Standards-cited register

Each module declares a standards-cited list as part of its output. The list identifies the standards, codes of practice and reference works underpinning that module's calculations. The deliverable's standards register, presented in Section 6, is the union of these lists across all modules invoked for the analysis.

4.3 Golden test case regime

Three reference site profiles — a 5 MW dairy, an 8 MW brewery and a 12 MW soft drinks plant — are maintained as fixtures alongside the engine source. Each fixture carries a golden truth block that records hand-checked engineering expectations against which each module's output is validated. The expected ranges, shortlist inclusions and exclusions, regulatory flags and characteristic outputs are calibrated against published sector benchmarks and cross-checked by a senior reviewer; they are not regenerated from engine output. A test that passes only because the engine has been run is not a test, and the regime is constructed to make this distinction explicit.

The numeric tolerance bands applied are ±10% on cardinal results and ±25% on probabilistic ranges (P10 / P90). Categorical results (shortlist membership, regulatory flag presence, warning codes) are checked for exact match.

4.4 Self-critique loop

The orchestration layer, before producing a final deliverable, re-reads the draft against a structured checklist: numerical consistency between sections, standards citations resolving to documents in the corpus, claims supported by tool calls, assumptions explicitly declared. Discrepancies are addressed by re-running the relevant engine modules, not by edit of the output text.

4.5 Boundary of machine-generated content

The deliverable is positioned as a draft for senior-engineer review. The system does not replace the chartered-engineer review and sign-off step. The boundary is explicit: numbers and citations are machine-generated; engineering judgement, sign-off and professional indemnity remain with the human reviewer.

5. Limitations and assumptions

The methodology, in its present version, applies to industrial sites within the food and drink, chemicals, pharmaceutical, paper and refining sectors, with thermal demand in the 1–50 MW range, considering the electrification of steam systems through heat pumps, electrode boilers and thermal storage, with optional photovoltaic, battery and waste-heat recovery integration. Sites outside this envelope — thermal demand below 1 MW or above 50 MW, sectors with materially different process characteristics (cement, glass, primary metals), or decisions other than steam-system electrification (cooling-system electrification, on-site hydrogen production, carbon capture) — fall outside the present validation envelope.

Default values, where used, are drawn from UK-specific datasets: DEFRA emission factors for the year of analysis; National Energy System Operator Future Energy Scenarios for grid carbon intensity projection; CIBSE benchmarks for sector load shapes; manufacturer-published data for equipment performance. Where a default is invoked, the provenance record discloses it; default values are visible to the reviewer and may be replaced with site-specific data without re-running the entire analysis.

The methodology does not perform detailed engineering design (line-sized P&IDs, foundation design, control narratives, hazardous-area drawings, single-line diagrams), nor does it conduct a HAZOP, a Quantitative Risk Assessment, a formal harmonic study, or a Distribution Network Operator connection study. Each of these is a downstream design activity for which the present analysis provides input, not substitute. The deliverable identifies, in each case, the activity required at the next design stage.

6. Standards register

The following register consolidates the standards, codes of practice and reference works cited across the methodology. A given Engine output cites only the subset relevant to the modules invoked for that analysis.

Reference	Title / scope
BS EN 14825	Air conditioners, liquid chilling packages and heat pumps — testing and rating at part-load conditions
BS EN 14511	Air conditioners, liquid chilling packages and heat pumps for space heating and cooling — performance
BS EN 16247-1	Energy audits — General requirements

BS EN 16247-3	Energy audits — Processes
BS EN 378 (1–4)	Refrigerating systems and heat pumps — Safety and environmental requirements
BS EN 50160	Voltage characteristics of electricity supplied by public distribution networks
BS EN 60300	Dependability management
BS 7671	Requirements for electrical installations (IET Wiring Regulations)
CIBSE AM17	Heat pumps in buildings (adapted for industrial application)
CIBSE TM54	Evaluating operational energy performance
Approved Document L	Conservation of fuel and power (Building Regulations, England)
CDM Regulations 2015	Construction (Design and Management) Regulations
DSEAR 2002	Dangerous Substances and Explosive Atmospheres Regulations
ATEX 2014/34/EU	Equipment for use in potentially explosive atmospheres
F-Gas Reg. 517/2014	Fluorinated greenhouse gases (UK retained)
Pressure Equipment (Safety) Regulations 2016	Implementing PED 2014/68/EU in the UK
PSSR 2000	Pressure Systems Safety Regulations
ENA EREC G99	Connection of generation to DNO networks
ENA EREC G5/5	Harmonic distortion limits at the point of common coupling
IEEE 519	Recommended practice for harmonic control in electric power systems
IEC 61078	Reliability block diagrams
GHG Protocol Corporate Standard	Greenhouse gas accounting and reporting
GHG Protocol Scope 2 Guidance	Location- and market-based methods
DEFRA Conversion Factors	UK Government GHG Conversion Factors (current year)
HM Treasury Green Book	Appraisal and evaluation methodology
NESO Future Energy Scenarios	UK grid carbon intensity projection (current year)
Kemp (2007)	Pinch Analysis and Process Integration, 2nd edition
Smith (2016)	Chemical Process Design and Integration, 2nd edition

Saltelli et al. (2008)	Global Sensitivity Analysis: The Primer
Lees (2012)	Loss Prevention in the Process Industries, 4th edition
OREDA (2015)	Offshore Reliability Data Handbook, 6th edition
IChemE	Process Integration Best Practice Guides

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